

Patrick Amadeus Irawan

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 Website  LinkedIn  Github

Research Interests

My research studies how multimodal systems ground vision, language, and action signals and uses the findings to build a general-purpose model that can perceive, reason, and act in complex real-world scenarios. My work spans post-training methods, architecture modification, and large-scale evaluation of multimodal & multilingual models within the vision-language domain.

Research Areas: computer vision, multimodal LLM, VLM, SFT, RL, evaluation.

Education

Ph.D. in Natural Language Processing — MBZUAI	2025 - 2029
· Advisors: Alham Fikri Aji and Yova Kementchedjhieva	
· <i>multimodal alignment; VLM; LLM; post-training; RL</i>	
Global Study Program — University of California, Davis	2023
· <i>computational neuroscience, tech management</i>	
Bachelor in Computer Science — Institut Teknologi Bandung	2020 - 2024
· <i>visual question answering; synthetic data generation; reasoning LLM</i>	

Publications

* **First author** † **Second author**

LinguDistill: Recovering Linguistic Ability in Vision Language Models via Selective Cross-Modal Distillation*	<i>Preprint</i> 2026
<i>VLM, SFT, post-training, knowledge distillation, architecture</i>	
Anthropogenic Regional Adaptation in Multimodal Vision-Language Model	<i>Under Review</i> 2026
<i>VLM, SFT, post-training, data curation, evaluation, multilingual, multimodal</i>	
Vision Language Models are Confused Tourists*	CVPR 2026
<i>VLM, visual reasoning, robustness, hallucination, interpretability</i>	
A Massive-Scale Multilingual Multi-Cultural Multimodal RAG†	CVPR 2026
<i>RAG, multilingual, multimodal, evaluation, test-time scaling</i>	
Counting to Four is Still a Chore for VLMs*	MMFM @ CVPR 2026
<i>VLM, numerical reasoning, interpretability, RL</i>	
Entropy2Vec: Crosslingual Language Modeling Entropy as End-to-End Learnable Language Representations*	MRL @ EMNLP 2025
<i>LLM, multilingual, representation learning, unsupervised learning</i>	
WorldCuisines: A Massive-Scale Benchmark for Multilingual and Multicultural VQA on Global Cuisines*	NAACL 2025 (Best Theme Paper 🏆)
<i>VQA, multilingual, multicultural, benchmark, evaluation, data curation</i>	
Seeing Culture: A Benchmark for Visual Reasoning and Grounding†	EMNLP 2025
<i>visual grounding, evaluation, object detection, multicultural</i>	
Predicting Language Model Performance on Multilingual Tasks via Proxy Models	NAACL 2025
<i>multilingual, evaluation, efficient, regression, SFT</i>	
Towards Efficient and Robust VQA-NLE Data Generation with Large Vision-Language Models*	COLING 2025
<i>VQA, VLM, NLP, synthetic data generation, visual reasoning</i>	
SEACrowd: A Multilingual Multimodal Data Hub and Benchmark Suite for Southeast Asian Languages	EMNLP 2024
<i>multilingual, multicultural, video understanding, speech understanding, evaluation</i>	

Work Experience

Research Engineer

Singapore Management University

Singapore

Feb 2025 – Sep 2025

- Led multimodal reasoning evaluation across 10K+ image-text samples, coordinating 5 researchers (juniors, externals), resulting in EMNLP main-track publication on VLM robustness for knowledge-intensive VQA and visual grounding.
- Designed and executed end-to-end data curation pipelines (data collection via crowdsourcing, filtering, packing), reducing experiment turnaround time and improving benchmarking coverage by 50%.

Software Engineer

IT Bauschmiede

Germany

Nov 2024 – Jul 2025

- Part of core logistics team optimizing real-time scheduling, invoicing, and geolocation systems serving 5+ enterprise clients, reducing manual operations by ~4%
- Architected microservices migration from monolithic MVC, reducing system latency by ~20% and improving system uptime and scalability

Data Scientist Intern

Supertype

2023

- Built predictive models reducing churn rate by 12% by replacing neural approaches with foundational statistical models. Improved model explainability and stakeholder adoption through feature-driven modeling.

Software Engineer Intern

Blibli

2022

- Scaled full-stack checkout and transaction testing coverage by 5% out of total volume over 6 months. Contributed majority on redundant SQL migration jobs project, reducing data pipeline overhead by 8% across services.

Software Engineer Intern

Ruangguru

2022

- Architected and scaled internal autograder system supporting 1K+ students/week, improving grading efficiency by ~70%

Achievement & Awards

OpenAI Micro Grant Award **Grantee**

2025

NAACL Best Theme Paper Award **Winner**

2025

BCG SEA Emeralds Case Competition **Winner**

2024

Indonesian International Student Mobility Awards @ UC Davis **Awardee**

2023

DOMO Higher Education Case Competition **Finalist**

2023

CSLeaders Full Bachelor Tuition Scholarship **Awardee**

2022

Gemastik XV Data Mining Division **Finalist**

2022

Advising & Mentorship

- **Rava Maulana** — world models and evaluation (*now Software Engineer at Tiket*)
- **Iskandar Muda** — world models and evaluation (*now M.Sc. student at Bandung Institute of Technology*)
- **Qinrong Cui** — test-time adaptation for fine-grained visual understanding (*now M.Sc. student at MBZUAI*)
- **Wilfried Ariel** — multicultural object detection and evaluation (*now Ph.D. student at NTU*)

References

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